

Student Number:									
-----------------	--	--	--	--	--	--	--	--	--

Surname:															
First Name(s):															

Department of Mathematics and Applied Mathematics
 MAM2000W: 2LA (Linear Algebra)

Test 2 Time : 1.5 hours	29 April 2019 Maximum marks: 51 Full marks: 50
--	---

- This question paper consists of 10 pages (including this one).
- Answer all questions in the spaces provided on this question paper. Use the backs of pages when you run out of space.
- Use your *UCT Examination Answer Book* for rough work. We won't take this work in! Only the answers that you write on this question paper will be marked.
- Calculators are not allowed. When your answer contains constants such as π , e or $\sqrt{2}$, leave them in that form. Don't simplify your answers.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.

DO NOT WRITE BELOW THIS LINE

Q1	Q2	Q3	Q4	Q5	Q6	Q7	TOTAL
13	8	6	10	5	4	5	51

Student Number:

--	--	--	--	--	--	--	--	--

Question 1. [13 points]

- a. Suppose A is a matrix for which $\det(A) = 3$ and $\det(2A) = 24$. Then

$$\det(AA^T) = \boxed{}, \quad \det(A^{-1}) = \boxed{},$$

$$\det(AA^T) = 9, \quad \det(A^{-1}) = 1/3,$$

and the dimensions of A are

--

$$3 \times 3.$$

Are the columns of A linearly independent? Give a reason for your answer.

--

Yes, because the determinant is non-zero.

- b. If $\det \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix} = 4$, then

$$\det \begin{pmatrix} b_1 & b_2 & b_3 \\ a_1 & a_2 & a_3 \\ -c_1 & -c_2 & -c_3 \end{pmatrix} = \boxed{}$$

$$4$$

and

$$\det \begin{pmatrix} 5a_1 + b_1 & 5a_2 + b_2 & 5a_3 + b_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix} = \boxed{}.$$

$$20$$

Student Number:								
-----------------	--	--	--	--	--	--	--	--

c. Let $B = \begin{pmatrix} 2 & 0 & -1 \\ 0 & -2 & 0 \\ -5 & -2 & -2 \end{pmatrix}$.

(i) Find all the eigenvalues of B .

(4)

For eigenvectors we need non-zero solutions to $B\mathbf{x} = \lambda\mathbf{x}$, i.e., $(\lambda I - B)\mathbf{x} = \mathbf{0}$, so we need

$$\begin{aligned} 0 &= \det(\lambda I - B) \\ &= \det \begin{pmatrix} \lambda - 2 & 0 & 1 \\ 0 & \lambda + 2 & 0 \\ 5 & 2 & \lambda + 2 \end{pmatrix} \\ &= (\lambda + 2) \det \begin{pmatrix} \lambda - 2 & 1 \\ 5 & \lambda + 2 \end{pmatrix} \\ &= (\lambda + 2)[(\lambda - 2)(\lambda + 2) - 5] \\ &= (\lambda + 2)[\lambda^2 - 9] \\ &= (\lambda + 2)(\lambda + 3)(\lambda - 3), \end{aligned}$$

which means the eigenvalues are $-2, -3, 3$.

(ii) Find all eigenvectors of B corresponding to the largest eigenvalue of B .

(3)

Eigenvectors are the non-zero solutions of $B\mathbf{x} = 3\mathbf{x}$, i.e., $(3I - B)\mathbf{x} = \mathbf{0}$, which has augmented matrix

$$\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 5 & 0 & 0 \\ 5 & 2 & 5 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 2 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

and solutions

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = s \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad s \in \mathbb{R}.$$

So the eigenvectors are the **non-zero** multiples of $\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$.

Student Number:								
-----------------	--	--	--	--	--	--	--	--

Question 2. [8 points]

- a. Let V be the real vector space of all real-valued functions on the interval $[0, 2]$. For each of the following subsets of V , say whether it is a subspace of V or not. Give full reasons for your answers.

(i) $U = \{f \in V : f(0) = f(2)\}.$ (5)

U contains the zero polynomial, and so it is not empty. Now let $f, g \in U$. Then

$$\begin{aligned}(f + g)(0) &= f(0) + g(0) \\ &= f(2) + g(2) \text{ since } f, g \in U \\ &= (f + g)(2),\end{aligned}$$

so $f + g \in U$. Similarly for $\lambda \in \mathbb{R}$,

$$\begin{aligned}(\lambda f)(0) &= \lambda f(0) \\ &= \lambda f(2) \text{ since } f \in U \\ &= (\lambda f)(2),\end{aligned}$$

so $\lambda f \in U$. Hence U is closed under vector addition and scalar multiplication, and is a subspace.

(ii) $W = \{f \in V : f(0) + f(1) = 1\}.$ (2)

W is not a subspace, because it does not contain the zero function.

OR because it is not closed under scalar multiplication (e.g., $f(x) = x \in W$ because $f(0) + f(1) = 0 + 1 = 1$ but $g(x) = 5x \notin W$ because $g(0) + g(1) = 0 + 5 = 5 \neq 1$).

OR because it is not closed under addition (if $f, g \in W$ then $(f + g)(0) + (f + g)(1) = (f(0) + f(1)) + (g(0) + g(1)) = 1 + 1 = 2 \neq 1$).

b. In $\mathbb{R}^3$, describe geometrically the subspace spanned by $\{(1, 2, 3)\}.$ (1)

It is a straight line through the origin. (MUST SAY through the origin.)

--	--	--	--	--	--	--	--	--

Question 3. [6 points] Let

$$\mathcal{T} = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \right\}.$$

a. Find $\text{span}(\mathcal{T})$. Give your answer in the form

$$\text{span}(\mathcal{T}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2 \times 2}(\mathbb{R}) : \dots \right\},$$

with the dots replaced by an equation containing only the variables a, b, c and d . (4)

$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{span}(\mathcal{T})$ if there exist $\lambda_1, \dots, \lambda_4$ such that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \lambda_1 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \lambda_3 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + \lambda_4 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

or

$$\lambda_1 + \lambda_3 + \lambda_4 = a$$

$$\lambda_2 = b$$

$$\lambda_2 = c$$

$$\lambda_1 - \lambda_3 = d.$$

which has augmented matrix

$$\begin{pmatrix} 1 & 0 & 1 & 1 & a \\ 0 & 1 & 0 & 0 & b \\ 0 & 1 & 0 & 0 & c \\ 1 & 0 & -1 & 0 & d \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 & 1 & a \\ 0 & 1 & 0 & 0 & b \\ 0 & 1 & 0 & 0 & c \\ 0 & 0 & -2 & -1 & d-a \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 & 1 & a \\ 0 & 1 & 0 & 0 & b \\ 0 & 0 & 0 & 0 & c-b \\ 0 & 0 & -2 & -1 & d-a \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 & 1 & a \\ 0 & 1 & 0 & 0 & b \\ 0 & 0 & -2 & -1 & d-a \\ 0 & 0 & 0 & 0 & c-b \end{pmatrix}$$

This has a solution for the λ_i if and only if $c = b$, so the span of $\mathcal{T}$ is

$$\text{span}(\mathcal{T}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2 \times 2}(\mathbb{R}) : c = b \right\}.$$

b. Is $\mathcal{T}$ a linearly independent subset of $M_{2 \times 2}(\mathbb{R})$? Give a reason for your answer. (2)

No. Note that if $a = b = c = d = 0$, then the system has infinitely many solutions. Therefore there are coefficients $\lambda_1, \dots, \lambda_4$, not all zero, such that

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \lambda_1 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \lambda_3 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + \lambda_4 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

and thus the system is linearly dependent.

OR No, because there are 4 vectors in $\mathcal{T}$ and $M_{2 \times 2}(\mathbb{R})$ has dimension 4. So if they were linearly independent then by a theorem they would be a basis for $M_{2 \times 2}(\mathbb{R})$, i.e., they would span the whole of $M_{2 \times 2}(\mathbb{R})$, which we showed in (a) they do not.

Student Number:								
-----------------	--	--	--	--	--	--	--	--

Question 4. [10 points] Read the theorem below and then answer the questions that follow.

Theorem: Let V be an n -dimensional vector space and $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a subset of V such that $\text{span}(S) = V$. Then S is a basis for V .

Proof: Suppose S is not a basis. (1)

Then there are scalars $\lambda_1, \lambda_2, \dots, \lambda_n$ not all zero such that

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n = \mathbf{0} \quad (2)$$

We may assume that $\lambda_1 \neq 0$. (3)

Let S_0 be the set obtained by removing $\mathbf{v}_1$ from S .

Then $\mathbf{v}_1 \in \text{span}(S_0)$. (4)

Hence $\text{span}(S_0) = \text{span}(S) = V$. (5)

But S_0 has $n - 1$ elements. (6)

This gives a contradiction. (7)

Hence S is a basis.

a. What method of proof is used here? (1)

Proof by contradiction.

b. How does the statement at (2) follow from the one at (2)? (1)

If $\text{span}(S) = V$ but S is not a basis, then S must be linearly dependent, which is what (2) says.

c. Why can we say, at (4), that $\mathbf{v}_1 \in \text{span}(S_0)$? (2)

If $\lambda_1 \neq 0$, then it follows from (2) that

$$\mathbf{v}_1 = -\frac{1}{\lambda_1}(\lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n);$$

hence $\mathbf{v}_1$ is a linear combination of the elements in S_0 and hence in $\text{span}(S_0)$.

Student Number:								
-----------------	--	--	--	--	--	--	--	--

- d. Since $S_0 \subset S$, we certainly have $\text{span}(S_0) \subseteq \text{span}(S)$. Explain why at (5) we can say $\text{span}(S_0) = \text{span}(S)$. (1)

Since $\mathbf{v}$ depends on other vectors in s , we can remove it from S without changing the span.

- e. What is the contradiction referred to at (7)? (You may need to quote a theorem here.) (2)

S_0 has $n - 1$ elements and spans V . But $\dim(V) = n$, and no set with fewer than n elements can span an n -dimensional vector space.

- f. Suppose R is a subset of V with $\text{span}(R) = V$, but that R is not a basis of V . What can be said about the number of elements in R ? Give a reason for your answer. (2)

The theorem proved here shows that R cannot have n elements (for then it would be a basis). But R cannot have fewer than n elements because it is a spanning set of V , so R must have more than n elements.

OR

Since R is a spanning set that is not linearly independent, it contains a basis, which has n elements, as a proper subset. Hence R must have more than n elements.

Student Number:								
-----------------	--	--	--	--	--	--	--	--

Question 5. [5 points] Prove the following theorem.

Theorem: Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be a basis for a vector space V and let $\mathbf{v}$ be a member of V . Then $\mathbf{v}$ can be expressed in exactly one way as a linear combination of vectors in S .

Since S spans V , we know $\mathbf{v}$ can be expressed as a linear combination of members of S . We need to show this can be done in only one way.

So suppose

$$\mathbf{v} = \lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \dots + \lambda_n \mathbf{u}_n = \mu_1 \mathbf{u}_1 + \mu_2 \mathbf{u}_2 + \dots + \mu_n \mathbf{u}_n.$$

Then

$$\begin{aligned} \mathbf{0} &= \mathbf{v} - \mathbf{v} = (\lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \dots + \lambda_n \mathbf{u}_n) - (\mu_1 \mathbf{u}_1 + \mu_2 \mathbf{u}_2 + \dots + \mu_n \mathbf{u}_n) \\ &= (\lambda_1 - \mu_1) \mathbf{u}_1 + (\lambda_2 - \mu_2) \mathbf{u}_2 + \dots + (\lambda_n - \mu_n) \mathbf{u}_n. \end{aligned}$$

But since S is a basis, the $\mathbf{u}_i$'s are linearly independent and so

$$\lambda_1 - \mu_1 = \lambda_2 - \mu_2 = \dots = \lambda_n - \mu_n = 0.$$

Therefore $\lambda_1 = \mu_1, \lambda_2 = \mu_2, \dots, \lambda_n = \mu_n$, which proves the result.

Student Number:								
-----------------	--	--	--	--	--	--	--	--

Question 6. [4 points] Let V be a vector space over $\mathbb{R}$, let $\mathbf{v} \in V$ and $\lambda \in \mathbb{R}$. Prove, using only the axioms for a vector space, that

a. $0\mathbf{v} = \mathbf{0}$. (2)

$$0\mathbf{v} = (0 + 0)\mathbf{v} = 0\mathbf{v} + 0\mathbf{v}$$

(by the distributive law), so

$$-(0\mathbf{v}) + 0\mathbf{v} = -(0\mathbf{v}) + [0\mathbf{v} + 0\mathbf{v}] = [-(0\mathbf{v}) + 0\mathbf{v}] + 0\mathbf{v}$$

(by the associative law) and therefore

$$\mathbf{0} = \mathbf{0} + 0\mathbf{v} = 0\mathbf{v}.$$

b. if $\lambda\mathbf{v} = \mathbf{0}$, then $\lambda = 0$ or $\mathbf{v} = \mathbf{0}$. (2)

Let $\lambda\mathbf{v} = \mathbf{0}$, and suppose $\lambda \neq 0$. Then

$$\mathbf{v} = 1\mathbf{v} = (\lambda^{-1}\lambda)\mathbf{v} = \lambda^{-1}(\lambda\mathbf{v}) = \lambda^{-1}\mathbf{0} = \mathbf{0},$$

(where we have used the fact that $\alpha\mathbf{0} = \mathbf{0}$ for every scalar α).

Student Number:								
-----------------	--	--	--	--	--	--	--	--

Question 7. [5 points] Let V be a finite-dimensional vector space. Let U and W be subspaces of V such that $U \cap W = \{\mathbf{0}\}$. Recall that

$$U + W = \{\mathbf{u} + \mathbf{w} : \mathbf{u} \in U, \mathbf{w} \in W\}.$$

Prove that $\dim(U + W) = \dim(U) + \dim(W)$.

Let $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k\}$ and $C = \{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_m\}$ be bases for U and W respectively.

Then

$$U = \left\{ \sum_{i=1}^k \lambda_i \mathbf{b}_i \mid \lambda_i \in \mathbb{R} \right\},$$

$$W = \left\{ \sum_{i=1}^m \mu_i \mathbf{c}_i \mid \mu_i \in \mathbb{R} \right\},$$

and so

$$U + W = \left\{ \sum_{i=1}^k \lambda_i \mathbf{b}_i + \sum_{i=1}^m \mu_i \mathbf{c}_i \mid \lambda_i, \mu_i \in \mathbb{R} \right\},$$

which shows $B \cup C$ is a spanning set for $U + W$.

To show $B \cup C$ is linearly independent, suppose

$$\lambda_1 \mathbf{b}_1 + \lambda_2 \mathbf{b}_2 + \dots + \lambda_k \mathbf{b}_k + \mu_1 \mathbf{c}_1 + \mu_2 \mathbf{c}_2 + \dots + \mu_m \mathbf{c}_m = \mathbf{0},$$

then

$$\underbrace{\lambda_1 \mathbf{b}_1 + \lambda_2 \mathbf{b}_2 + \dots + \lambda_k \mathbf{b}_k}_{\in U} = - \underbrace{(\mu_1 \mathbf{c}_1 + \mu_2 \mathbf{c}_2 + \dots + \mu_m \mathbf{c}_m)}_{\in W}.$$

Since $U \cap W = \{\mathbf{0}\}$, this implies that

$$\lambda_1 \mathbf{b}_1 + \lambda_2 \mathbf{b}_2 + \dots + \lambda_k \mathbf{b}_k = \mu_1 \mathbf{c}_1 + \mu_2 \mathbf{c}_2 + \dots + \mu_m \mathbf{c}_m = \mathbf{0}.$$

So (since B and C are linearly independent) $\lambda_1 = \dots = \lambda_k = \mu_1 = \dots = \mu_m = 0$, so $B \cup C$ is linearly independent and hence a basis for $U + W$. Thus $\dim(U + W) = k + m = \dim(U) + \dim(W)$.